

Feature Selection Report

Best Model: CatBoost (Run 7)

Reasons:

- Strong top variable score (0.591).
- Balanced and meaningful variable selections including transaction time windows and state-level aggregations.
- CatBoost is also known to perform better on tabular data with default settings compared to LGBM/RandomForest unless those are finely tuned.

Inferences from the Runs

1. Consistently High Filter Scores
 - Across all models, the variable Cardnum_unique_count_for_card_state_1 appears at the top with a filter score of ~0.519. This suggests it's a strong and stable feature across multiple model types.
2. Top-Performing Model by Individual Variable Score
 - In Run 7 (CatBoost), Cardnum_total_1 achieves the highest single filter score: 0.591, which is higher than any score from LGBM or Random Forest runs. This is a strong indication of CatBoost's effectiveness in identifying informative variables.
3. Model Sophistication and Feature Selection
 - CatBoost can handle categorical variables natively and often performs well with structured data like transaction logs.
 - Random Forest identified more diverse features but didn't surpass the CatBoost filter scores.

First Run – Baseline Model (LGBM)

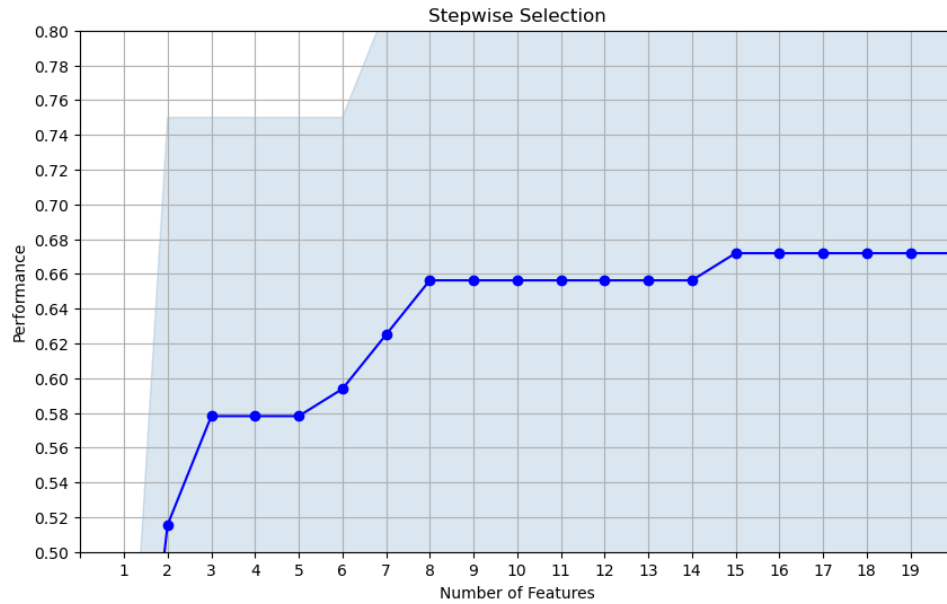
Model: LGBMClassifier

Wrapper Complexity: n_estimators = 5

Feature Selection: Forward

num_out_of_filter: 200

Wrapper Parameters: n_estimators = 20, num_leaves = 4



CPU times: user 2.95 s, sys: 273 ms, total: 3.22 s
Wall time: 1min 16s

List of top 20 variables:

	wrapper order	variable	filter score
0	1	U*_cardnum	0.436571
1	2	Cardnum_total_3	0.484935
2	3	Cardnum_max_14	0.371055
3	4	Cardnum_total_7	0.515563
4	5	Cardnum_count_14	0.464369
5	6	Cardnum_count_7	0.457883
6	7	card_zip_day_since	0.486312
7	8	Cardnum_total_14	0.400785
8	9	Cardnum_count_30	0.437424
9	10	Cardnum_count_60	0.437424
10	11	Cardnum_total_0	0.391105
11	12	Cardnum_day_since	0.399146
12	13	Cardnum_actual/max_0	0.406090
13	14	Cardnum_actual/toal_7	0.361752
14	15	card_state_day_since	0.263324
15	16	Cardnum_max_3	0.357243
16	17	Dow_Risk	0.344416
17	18	Cardnum_actual/med_1	0.334802
18	19	Cardnum_actual/med_0	0.333421
19	20	Cardnum_actual/med_3	0.323105

Second Run – LGBM (Backward)

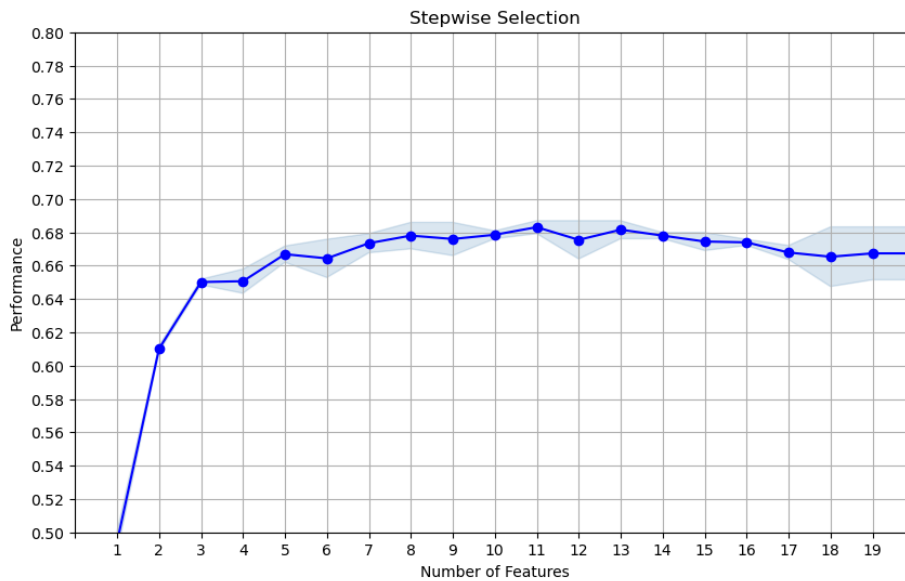
Model: LGBMClassifier

Wrapper Complexity: n_estimators = 5

Feature Selection: Backward

num_out_of_filter: 200

Wrapper Parameters: n_estimators = 20, num_leaves = 4



```
CPU times: user 54.4 s, sys: 12.9 s, total: 1min 7s
Wall time: 2h 46min 24s
```

► SequentialFeatureSelector ⓘ

► estimator: LGBMClassifier

► LGBMClassifier

List of top 20 variables:

Wrapper Order	Variable	Filter Score
1	Cardnum_unique_count_for_card_state_1	0.5186059951957085
2	card_zip_total_amount_1_by_60	0.3458850828428253
3	Card_dow_vdratio_0by60	0.50844776655556563
4	Cardnum_total_7	0.5159873870717485
5	Cardnum_total_30	0.3691095461455587

6	Cardnum_unique_count_for_card_zip_60	0.3014462545628256
7	card_merch_vdratio_1by30	0.2733331094751868
8	card_merch_vdratio_0by30	0.2733331094751868
9	card_merch_day_since	0.2733706479521256
10	card_merch_vdratio_0by7	0.2733706479521257
11	card_merch_vdratio_1by14	0.2733706479521257
12	card_merch_vdratio_1by7	0.2733706479521257
13	card_merch_vdratio_0by14	0.2733706479521257
14	card_zip_count_1_by_30_sq	0.2737839069856730
15	Card_dow_unique_count_for_state_des_60	0.2747260118836998
16	Card_dow_unique_count_for_Card_Merchdesc_60	0.2747760631862850
17	card_zip_vdratio_0by60	0.2762725401131096
18	card_zip_vdratio_1by60	0.2762725401131096
19	card_zip_vdratio_0by30	0.2764352068465113
20	card_zip_vdratio_1by30	0.2764352068465113

Third Run – Random Forest

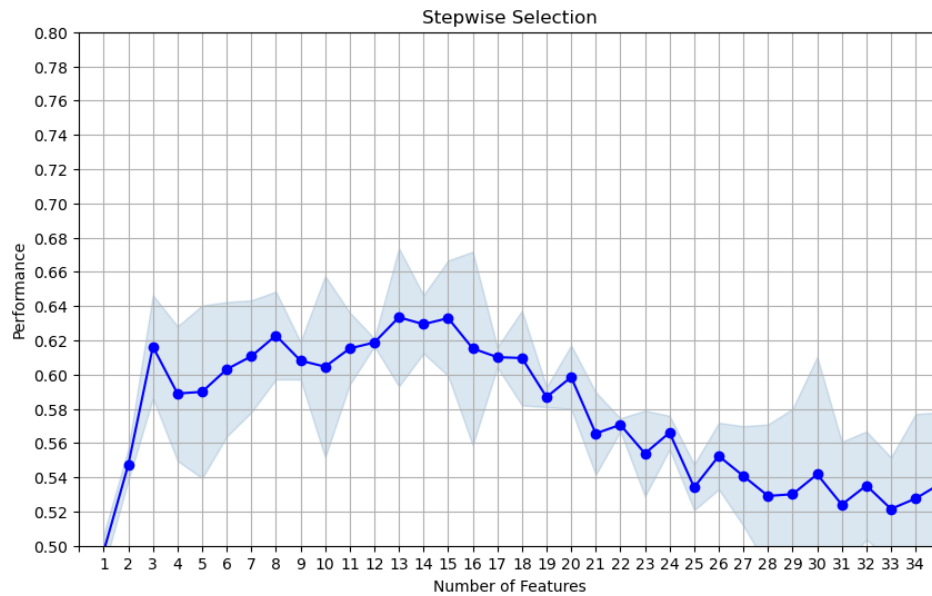
Model: RandomForestClassifier

Wrapper Complexity: n_estimators = 5

Feature Selection: Forward

num_out_of_filter: 400

num_out_of_wrapper: 35



```
CPU times: user 20.6 s, sys: 5.16 s, total: 25.8 s
Wall time: 17min 58s
```



List of top 20 variables:

Wrapper Order	Variable	Filter Score
1	Cardnum_unique_count_for_card_state_1	0.5186059951957085
2	Card_Merchnum_desc_vdratio_1by60	0.2700875169200329
3	Card_Merchnum_desc_total_30	0.2543774394070377
4	card_merch_total_amount_0_by_60	0.2357413374024831
5	Card_Merchnum_Zip_count_1_by_30	0.2450144626089843
6	Card_Merchdesc_Zip_vdratio_0by60	0.2777984459387328
7	card_merch_count_1_by_60_sq	0.3045681000240437
8	Card_Merchnum_Zip_vdratio_1by60	0.2725814592869083
9	Card_Merchnum_Zip_total_amount_1_by_60	0.3058974499494893
10	Card_Merchnum_Zip_count_1_by_60	0.3039107363528629
11	Card_Merchnum_desc_vdratio_0by60	0.2700875169200329
12	Card_Merchnum_Zip_count_0_by_60_sq	0.2359292135198072
13	Card_dow_unique_count_for_Card_Merchdesc_1	0.4868859821823698
14	Card_Merchdesc_Zip_vdratio_1by60	0.2777984459387328
15	Card_Merchdesc_total_3	0.2598959439754863
16	Card_Merchdesc_State_vdratio_0by60	0.2788128654655833
17	Card_Merchnum_Zip_total_amount_1_by_30	0.2493830163259113
18	Cardnum_count_0_by_14	0.2444695243003826
19	Merchdesc_Zip_total_3	0.2521105462212693
20	Card_dow_total_amount_0_by_60	0.3268602849566375

Fourth Run – Random Forest

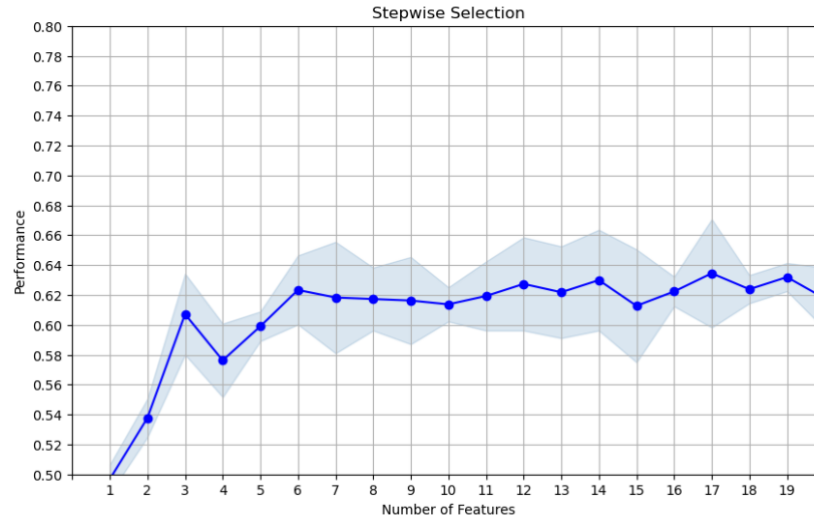
Model: RandomForestClassifier

Wrapper Complexity: n_estimators = 5

Feature Selection: Forward

num_out_of_filter: 400

num_out_of_wrapper: 20



```
CPU times: user 10.3 s, sys: 2.59 s, total: 12.9 s
Wall time: 6min 49s
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SequentialFeatureSelector ⓘ
  estimator: RandomForestClassifier
    RandomForestClassifier ⓘ
```

List of top 20 variables:

Wrapper Order	Variable	Filter Score
1	Cardnum_unique_count_for_card_state_1	0.5186059951957085
2	Card_Merchdesc_vdratio_1by60	0.2791824341468994
3	Card_Merchnum_desc_total_60	0.2407418173247832
4	Card_Merchnum_State_vdratio_0by60	0.2729702375314444
5	Card_Merchdesc_Zip_vdratio_1by60	0.2777984459387328
6	Card_Merchnum_State_count_1_by_60	0.3043929204649957
7	Card_Merchnum_State_count_1_by_30	0.2450211593465581
8	Card_Merchdesc_vdratio_0by60	0.2791824341468994
9	Card_Merchnum_State_total_amount_0_by_60	0.2355661578434351
10	card_merch_vdratio_1by60	0.2732330068700165
11	card_merch_total_amount_1_by_30	0.2489092902503547
12	Merchnum_total_1	0.2431599668140857
13	Card_Merchnum_Zip_total_amount_1_by_60	0.3058974499494893
14	Card_Merchnum_Zip_count_1_by_30	0.2450144626089843
15	card_merch_count_1_by_60_sq	0.3045681000240437
16	Card_Merchnum_State_count_1_by_30_sq	0.2450211593465581
17	Card_Merchdesc_Zip_vdratio_0by60	0.2777984459387328
18	Card_Merchnum_State_total_amount_1_by_30	0.2487841619938918
19	Card_Merchdesc_total_1	0.2436526997134087
20	Card_dow_unique_count_for_Card_Merchdesc_1	0.4868859821823698

Fifth Run – Random Forest

Model: RandomForestClassifier

Wrapper Complexity: n_estimators = 5

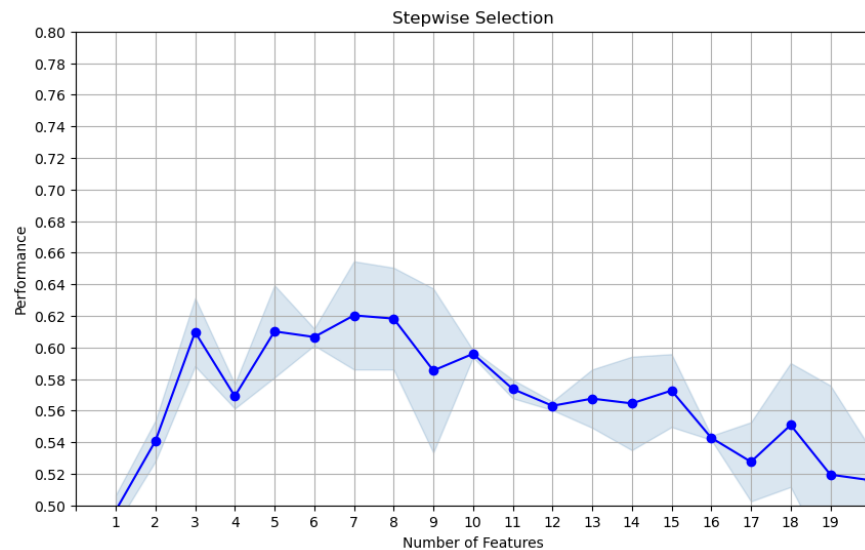
Feature Selection: Forward

num_out_of_filter: 350

num_out_of_wrapper: 20

```
CPU times: user 9.2 s, sys: 2.35 s, total: 11.5 s
Wall time: 7min 29s

> SequentialFeatureSelector ⓘ
> estimator: RandomForestClassifier
  > RandomForestClassifier ⓘ
```



List of top 20 variables:

Wrapper Order	Variable	Filter Score
1	Cardnum_unique_count_for_card_state_1	0.5186059951957085
2	Card_Merchnum_desc_vdratio_1by60	0.2700875169200329
3	Card_Merchnum_Zip_total_3	0.2546391443634789
4	Card_Merchdesc_vdratio_0by60	0.2791824341468994
5	Card_Merchnum_desc_vdratio_0by60	0.2700875169200329
6	Card_Merchdesc_State_vdratio_0by60	0.2788128654655833
7	Merchnum_total_1	0.2431599668140857
8	Card_Merchdesc_State_vdratio_1by60	0.2788128654655833
9	Card_Merchnum_State_total_30	0.2539895418120028
10	Card_dow_count_0_by_60_sq	0.3738634822211754
11	Card_Merchdesc_total_30	0.2552448791657018

12	Card_Merchnum_State_count_0_by_60	0.2364364232832324
13	card_zip_total_amount_1_by_30	0.2810112010380260
14	Card_dow_unique_count_for_state_des_7	0.4232198256377344
15	Cardnum_unique_count_for_card_state_14	0.4416929948133546
16	Card_Merchnum_desc_total_60	0.2407418173247832
17	Card_dow_count_0_by_60	0.3738634822211754
18	Cardnum_max_3	0.4662718780420817
19	Cardnum_actual/max_0	0.4609703807161835
20	card_zip_max_1	0.2400758879116918

Sixth Run – CatBoost

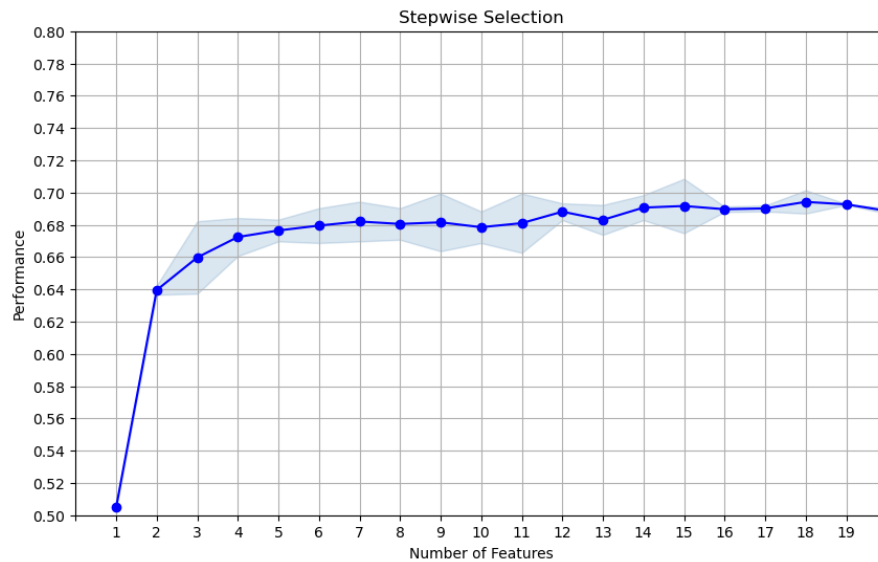
Model: CatBoostClassifier

Wrapper Complexity: iterations = 20, depth = 3

Feature Selection: Forward

num_out_of_filter: 200

num_out_of_wrapper: 20



```
CPU times: user 5.68 s, sys: 1.78 s, total: 7.46 s
Wall time: 7min 17s
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> SequentialFeatureSelector ⓘ
  estimator: CatBoostClassifier
    > CatBoostClassifier

```

List of top 20 variables:

Wrapper Order	Variable	Filter Score
1	Cardnum_unique_count_for_card_state_1	0.5186059951957085
2	Cardnum_total_1	0.5911137143420107
3	Card_dow_count_0_by_60	0.3738634822211754
4	card_zip_total_amount_1_by_30	0.2810112010380260
5	Cardnum_vdratio_0by14	0.4017270867195838
6	Card_Merchnum_Zip_total_amount_1_by_60	0.3058974499494893
7	card_state_total_amount_1_by_60	0.3488908916449487
8	Cardnum_avg_0	0.3388579940008128
9	Card_dow_max_14	0.4319636171379462
10	Cardnum_total_60	0.2876972314343721
11	Card_dow_actual/max_7	0.3596461246195071
12	Cardnum_actual/toal_3	0.4156378832607220
13	card_state_total_3	0.2915713416378370
14	card_zip_count_1_by_30	0.2737839069856730
15	card_state_count_1_by_60_sq	0.3443984197726720
16	Card_dow_actual/toal_7	0.3941931555794056
17	Cardnum_max_7	0.3799071073166451
18	card_state_count_1_by_30_sq	0.3048364256907635
19	Cardnum_total_amount_0_by_60	0.3702963005435002
20	card_state_total_amount_1_by_30	0.2998429086015698

Seventh Run – CatBoost

Model: CatBoostClassifier

Wrapper Complexity: iterations = 20, depth = 3

Feature Selection: Forward

num_out_of_filter: 300

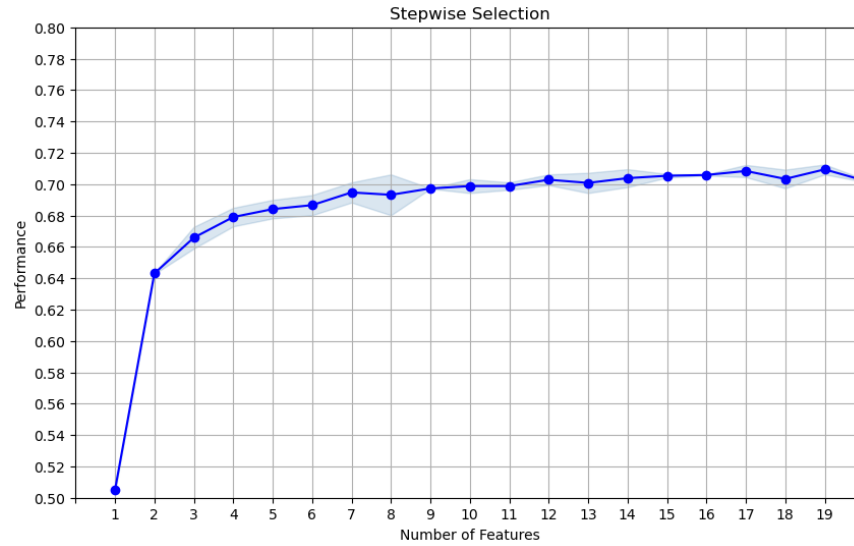
num_out_of_wrapper: 20

```
CPU times: user 8.98 s, sys: 2.83 s, total: 11.8 s
Wall time: 11min 6s
```

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> SequentialFeatureSelector ④
  > estimator: CatBoostClassifier
    > CatBoostClassifier

```



List of top 20 variables:

Wrapper Order	Variable	Filter Score
1	Cardnum_unique_count_for_card_state_1	0.5186059951957085
2	Card_Merchdesc_State_total_3	0.2599209696267789
3	Card_Merchnum_Zip_total_amount_1_by_60	0.3058974499494893
4	Cardnum_max_0	0.4923163140953655
5	card_merch_total_7	0.2598647537776855
6	Cardnum_total_amount_0_by_60	0.3702963005435002
7	card_zip_total_amount_0_by_60	0.2655482819256827
8	card_state_max_7	0.2539558237066410
9	Card_dow_vdratio_0by14	0.5026571349557347
10	Card_dow_total_30	0.4268030110610211
11	High_Amount_For_Merchant	0.2515712466037181
12	Cardnum_variability_med_0	0.2691895078531444
13	Card_dow_avg_60	0.2510064905135359
14	card_state_total_14	0.2642250521974898
15	Card_dow_actual/toal_14	0.3370331741934217
16	Card_dow_unique_count_for_Card_Merchdesc_1	0.4868859821823698
17	Card_dow_count_0_by_60_sq	0.3738634822211754
18	Merchnum_desc_State_total_3	0.2491566197124204
19	Cardnum_unique_count_for_card_state_60	0.3273076232251982
20	Cardnum_variability_max_3	0.4159074760490262